

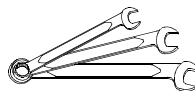
SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☐	Engage turning gear
☒	Shut off cooling water
☐	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

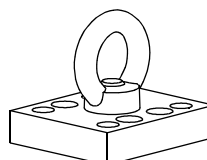
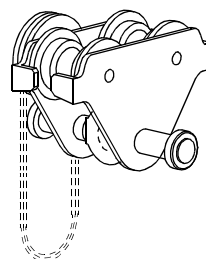
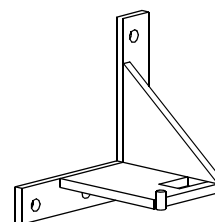
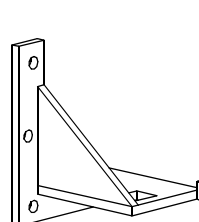
Data

Ref.	Description	Value	Unit
D-1	Air cooler element	1550	kg

Standard Tools: See Section 913



2 x 2000 kg





Spare Parts

Plate – Item No.	Description	Qty
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1. For the day-to-day checking of the scavenge air cooler, measure the drop in pressure and temperature of the scavenge air across the cooler while the engine is running.

Compare these measurements with the testbed data. *For further evaluation of the measurements, see instruction book Volume I, Chapter 706.*

2. An increase in the air pressure drop across the scavenge air cooler indicates fouling of the air side of the cooler.

A decrease in the air temperature drop across the scavenge air cooler indicates fouling of the water side of the cooler.

3. If the measurements indicate fouling, it is recommended to inspect the air cooler elements as follows:

4. Remove the front end cover of the air cooler.

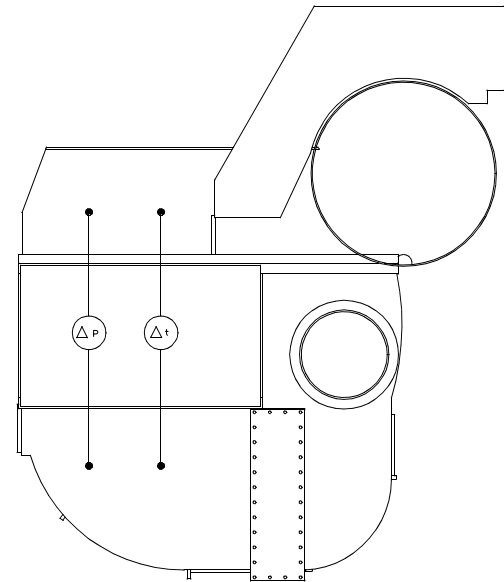
5. Visually inspect the air cooler element through the opening.

Clean the element if it is fouled.

See Procedure 910-1.3.

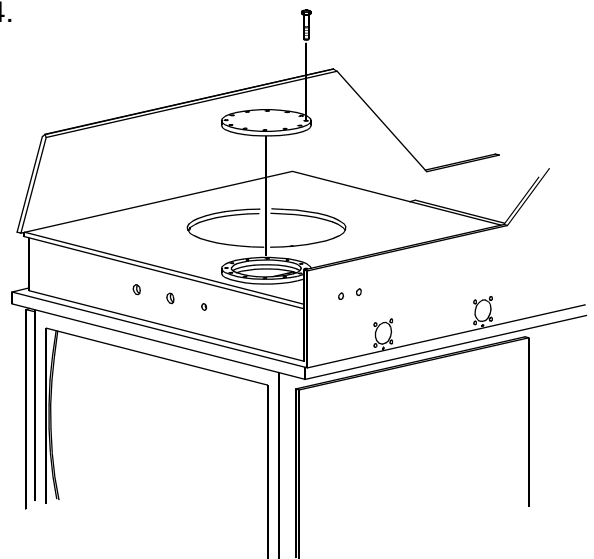
If the element needs to be dismantled for further inspection or replacement, *see Procedure 910-1.2.*

1.



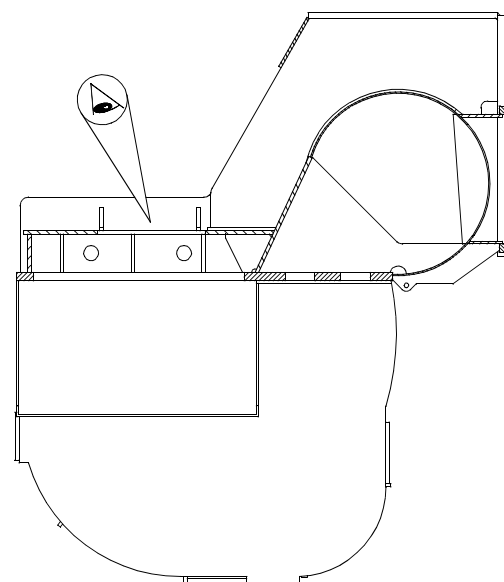
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4.

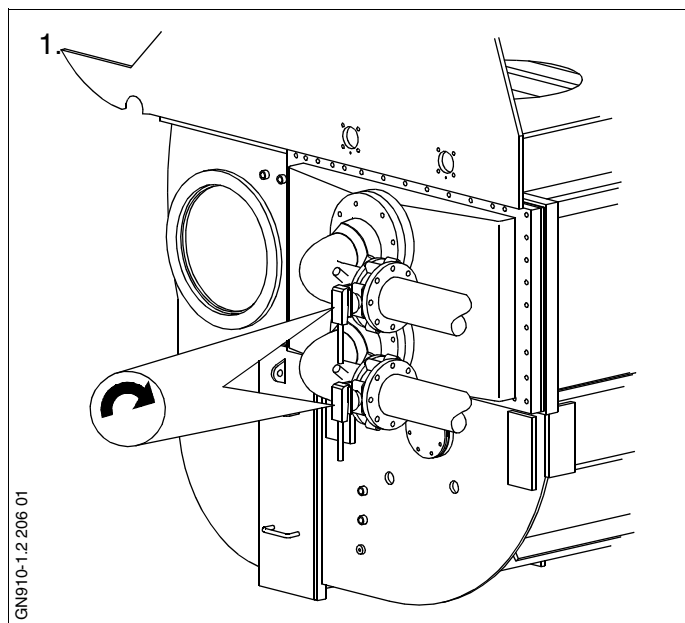


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5.



GN910-1.1 206 06



The cooler element normally does not need to be dismantled for inspection purposes. It is only in the event that the cooler element has suffered damage that it has to be dismantled and replaced.

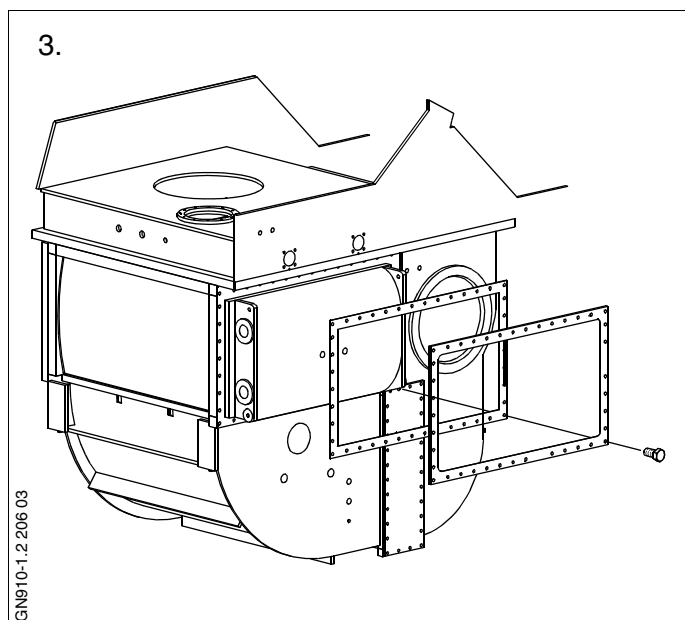
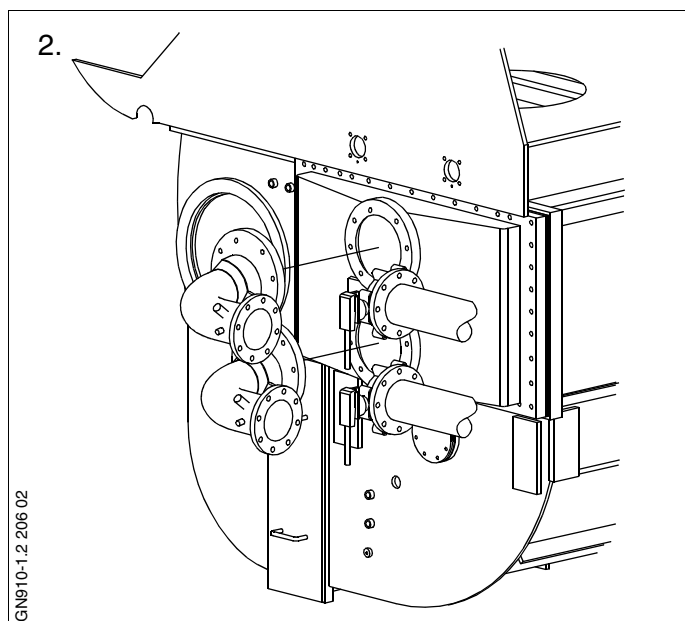
Note!

During dismantling, take care not to damage the cooler element.

1. Close the cooling water inlet and outlet valves. Open the drain cocks in the aft end cover to drain off the cooling water.
2. Dismount the cooling water inlet and outlet pipes from the aft end cover.
3. Release the screws and remove the frame around the front end cover.

Remove and discard the O-ring.

Remove the sealing plate.



4. Mount the first set of guide rails on the frame around the front end cover.
5. When mounting the guide rails, hold them up against the wheels of the cooler element and mount, but do not tighten, the lowermost screws. Mount and tighten the uppermost screws.

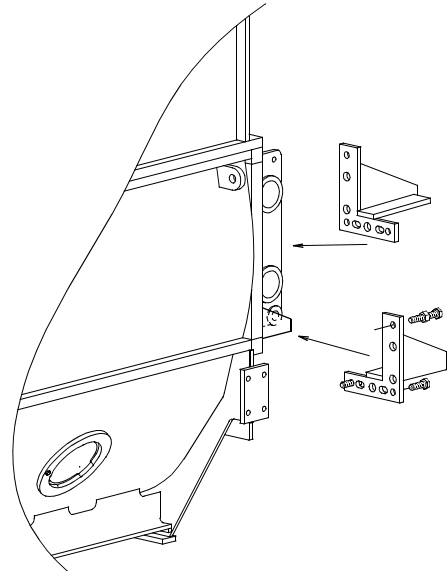
Finally, tighten the lowermost screws and the thrust nuts on the uppermost screws simultaneously. This results in loosening and lifting the cooler element.

6. Mount the lifting bracket on the aft end cover.

Mount a travelling trolley on the beam above the aft end cover.

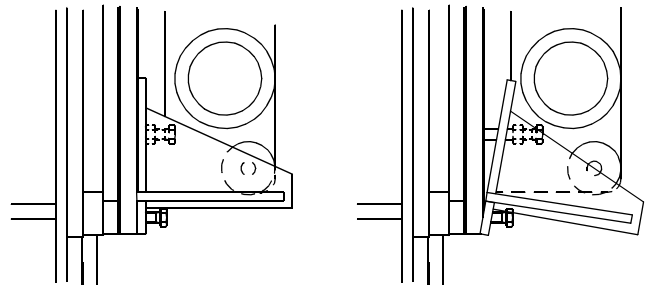
Mount a tackle between the travelling trolley and the lifting bracket, and use the tackle to lift the air cooler element approx. 1-2 mm.

4.



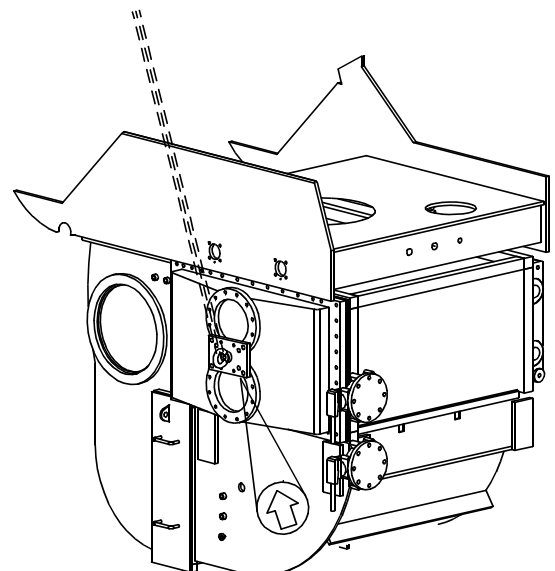
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5.



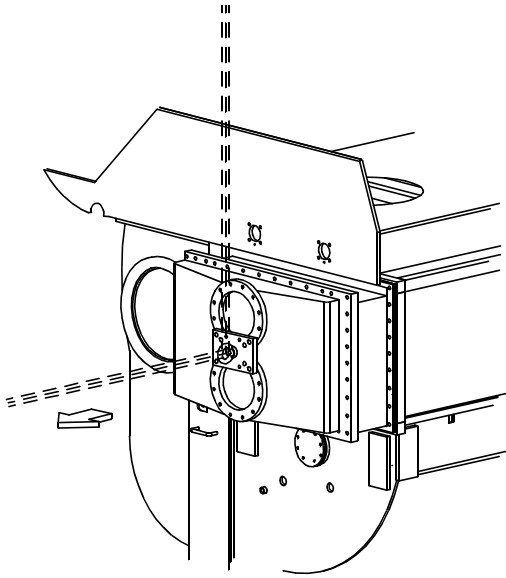
GN910-1.2 206 05

6.



GN910-1.2 228 06

7.



7. Mount a second tackle on the lifting bracket, and use it to pull out the air cooler element approx. half a metre.

8. Mount the second set of guide rails on the frame around the air cooler element.

9. Pull out the air cooler element until the front end cover of the air cooler element is clear of the air cooler housing.

Mount a second travelling trolley on the beam above the air cooler.

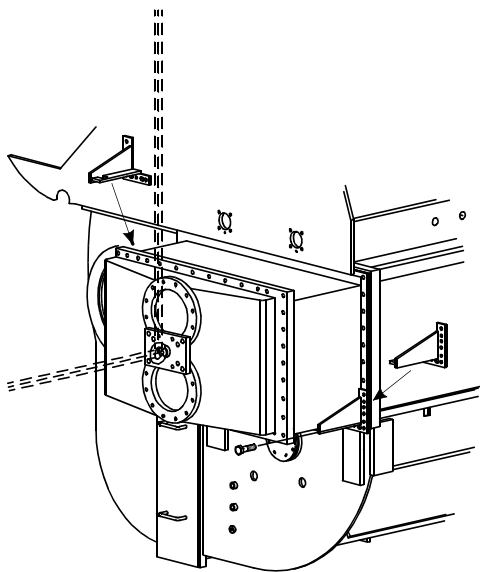
Mount a tackle between the second travelling trolley and the lifting holes in the upper corners of the front end cover of the air cooler element.

Using the tackles, lift the air cooler element away and land it on a couple of wooden planks.

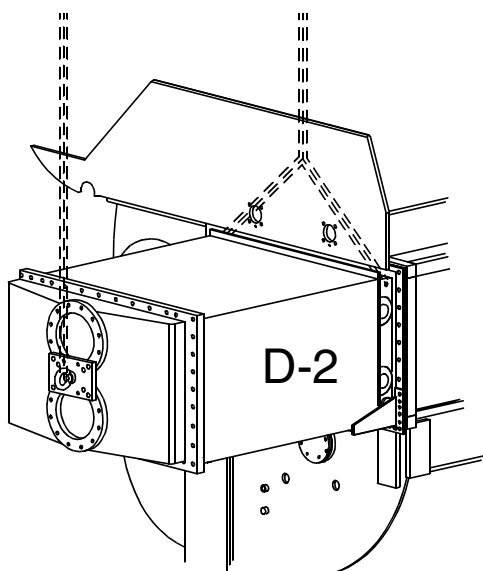
Note!

Depending on the engine layout and engine room installation, it may be necessary to remove some pipes or floor plates.

8.



9.



Cleaning the air side:

The air side of the cooler is cleaned by injecting a chemical fluid through the spray pipe arrangement fitted to the air chamber above the cooler element.

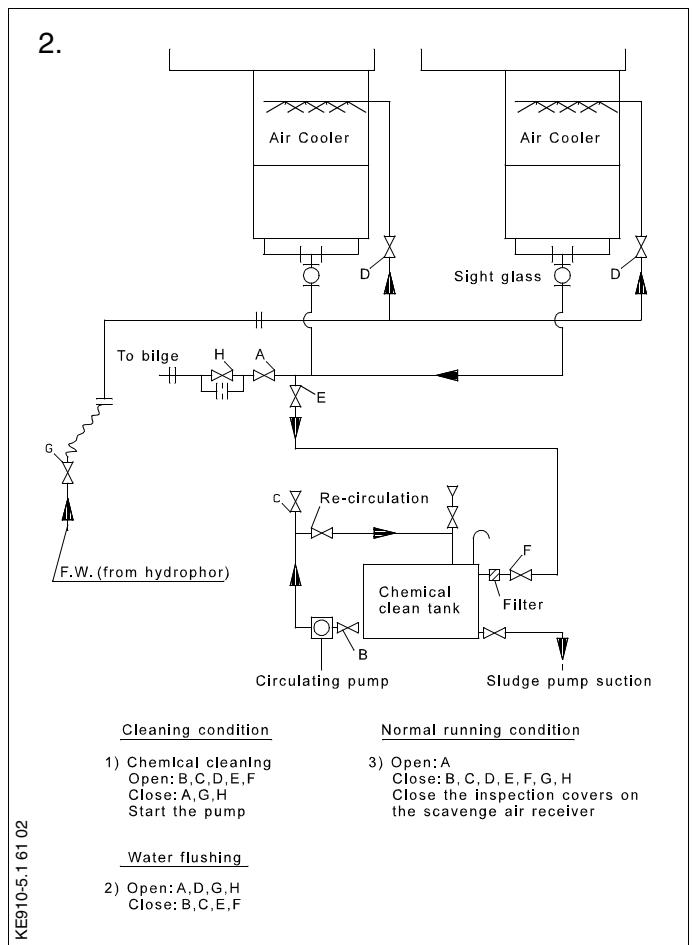
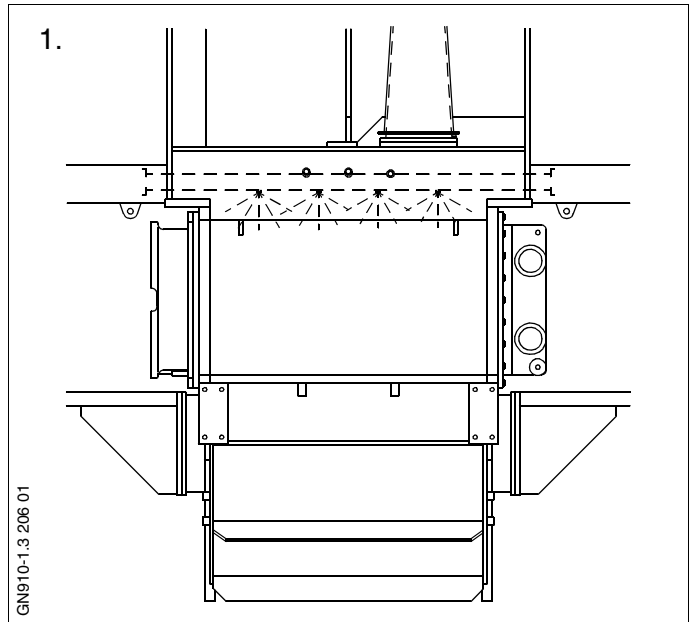
It is recommended to use one of the following cleaning fluids, or a similar product.

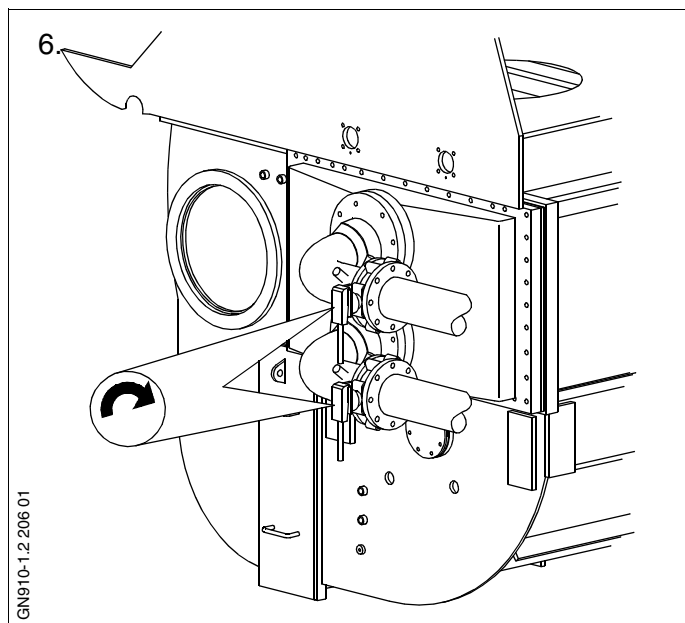
1) Product: ACC 9, produced by Drew Chemical Corp., New York, USA

2) Product: 80B, produced by Vecom Int., Maassluis, Holland

Cleaning should be carried out in the following sequence:

1. Do not start cleaning until the engine has been at a standstill for about 30 minutes. Do not disconnect the compressed air supply to the exhaust valve.
2. Follow the detailed cleaning instructions displayed at the cleaning pipe on the engine. To ensure satisfactory spraying of the cleaning fluid, the circulating pump pressure must be at least 0.7 bar.
3. Continue the cleaning process for at least 30 minutes. The time required depends on the frequency with which cleaning is carried out and on the chemical product used.
4. After cleaning, flush the cooler with clean water until the water appearing in the sight glasses is clean and pure.
5. Inspect the element either by removing the cover on top of the cooler or by dismantling the charging air pipe.



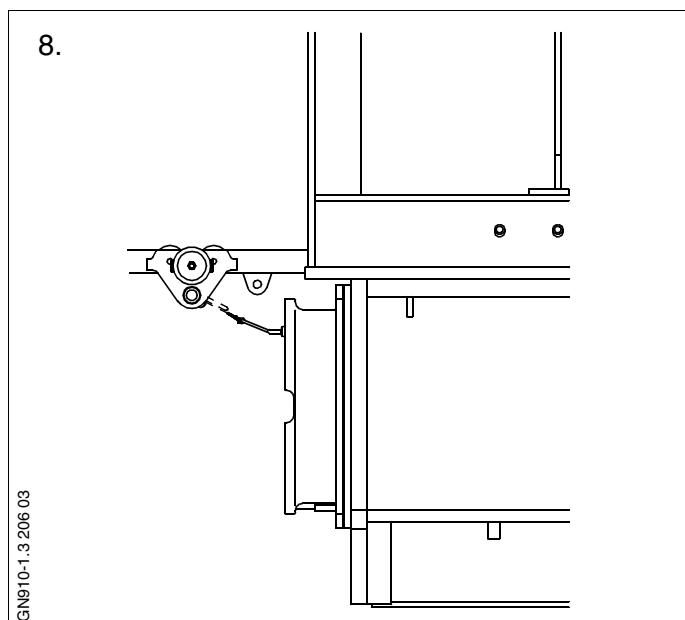
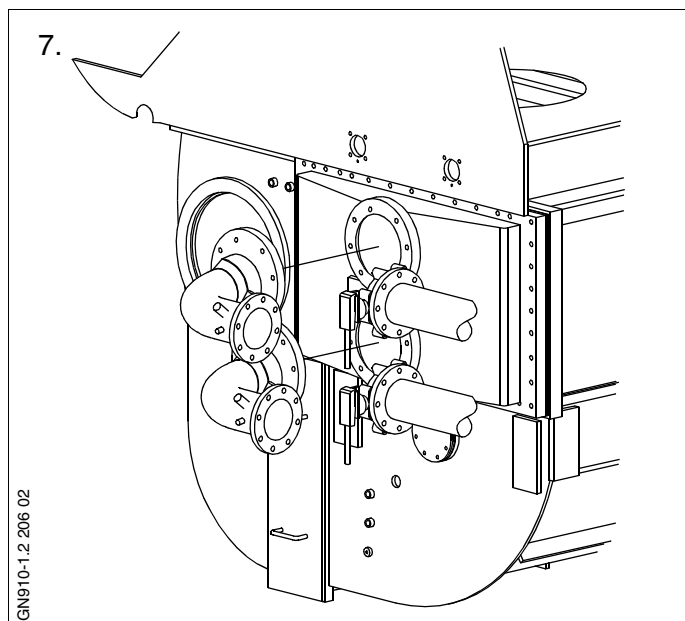


Cleaning the water side

6. Close the cooling water inlet and outlet valves. Remove the plug screws from the front end cover and the aft end cover to drain off the cooling water.
7. Dismount the cooling water inlet and outlet pipes from the aft end cover.
8. Screw two eye bolts into the upper flange of the aft end cover. Mount a lifting wire rope between the eye bolts.

Mount a travelling trolley on the beam above the aft end cover.

Mount a tackle between the travelling trolley and the lifting wire rope. Tighten up the tackle.



9. Mount a shackle in each of the two lifting eyes on the front end cover.

Mount two tackles above the front end of the air cooler.

Hook the tackles on to the shackles and tighten up.

10. Remove the screws of the front end and aft end covers.

Lower both covers and land them on a couple of wooden planks. Discard the gaskets.

11. Clean the inside of the tubes, using the cleaning brush mounted on a drilling machine. In the event of leakages between cooling tube and tube plate, the tubes are rolled with the tube expander.

Tools for tube cleaning and the tube expander are supplied by the air cooler manufacturer.

Clean the front-end and aft-end covers inside with a stiff brush and then rinsed with fresh water.

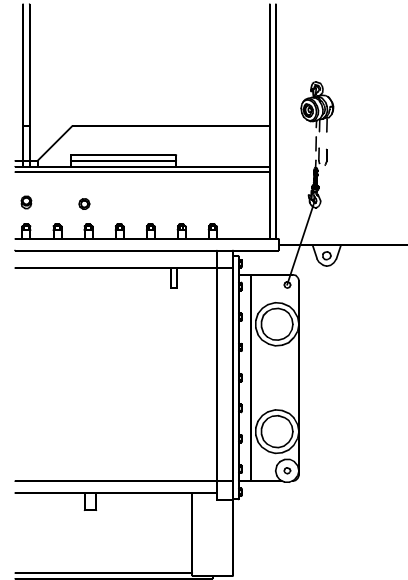
Replace the anti-corrosion blocks, if necessary.

Note!

If damaged, the coating inside the covers is to be repaired with the original coating or a similar product.

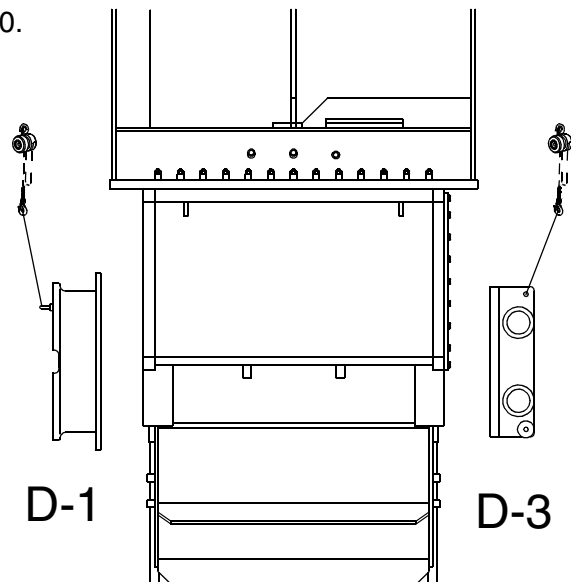
12. When cleaning is complete, mount the covers in the reverse order to dismantling.

9.



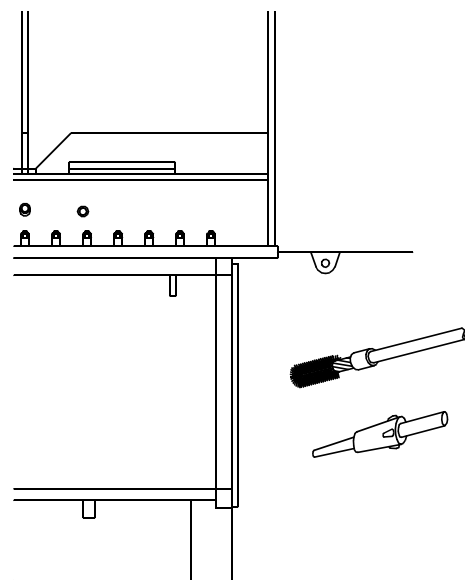
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10.

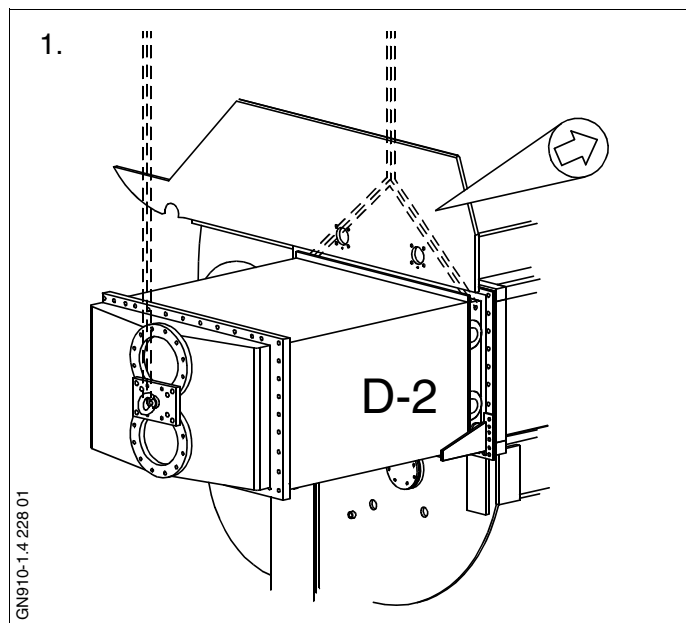


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11.



GN910-1.3 206 06



1. Lift the air cooler element and land the front end of the element on the guide rails mounted on the aft end of the air cooler housing.

Remove the tackle from the front end of the cooler.

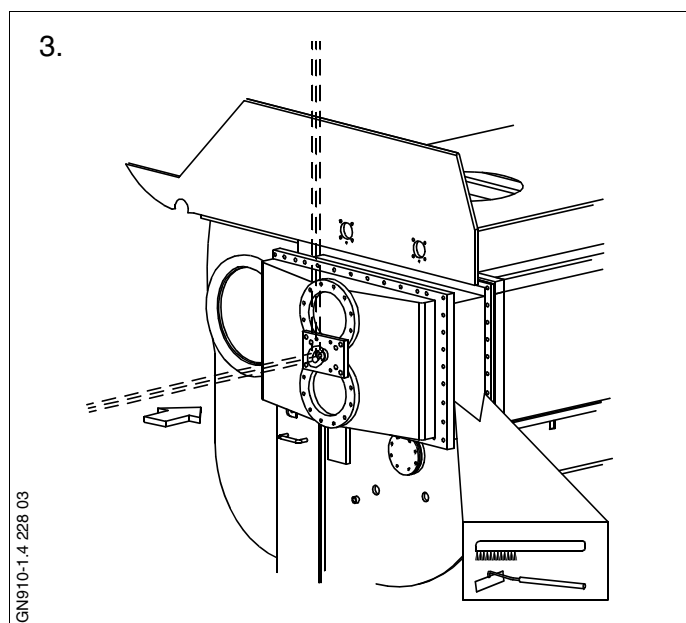
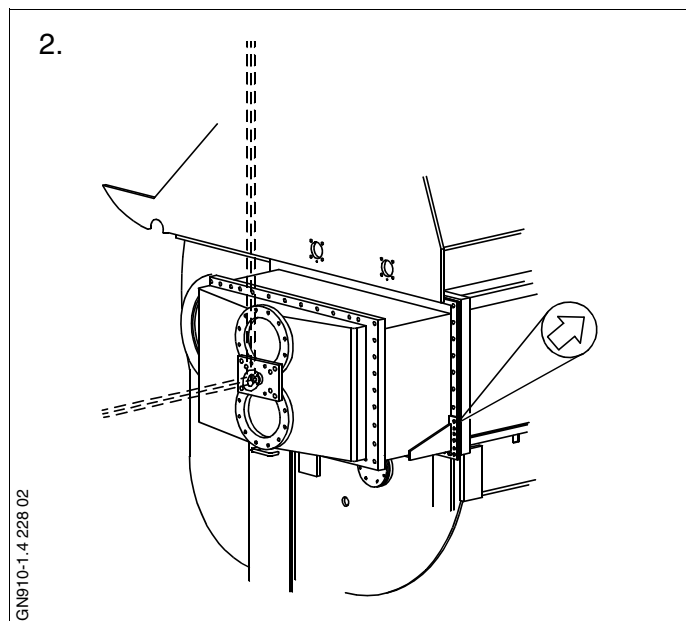
2. Push the air cooler element inwards approx. one metre.

Remove the guide rails from the aft end of the air cooler housing.

3. Clean the frame around the air cooler element and the contact surface of the aft end cover.

Apply a thin layer of silicone paste to the frame around the air cooler element and the contact surface of the aft end cover.

Push the air cooler element fully into the air cooler housing.



4. Mount and tighten all the screws of the aft end cover.

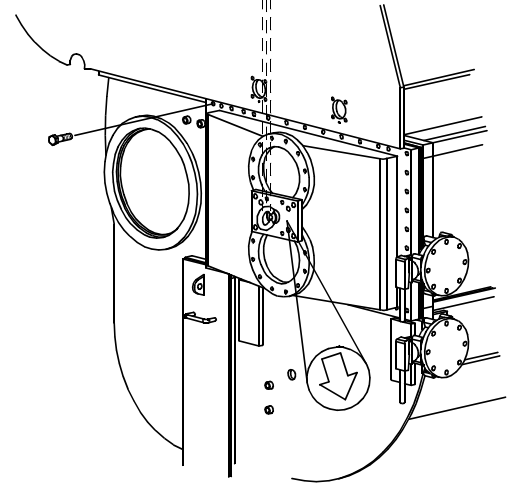
Remove the tackle and the lifting bracket from the aft end cover.

5. Remove the guide rails from the front end cover.
6. Mount the sealing plate, a new O-ring and the frame around the front end cover.
7. Mount the cooling water inlet and outlet pipes on the aft end cover.
8. Close the drain cocks, and open the cooling water inlet and outlet valves.

Note!

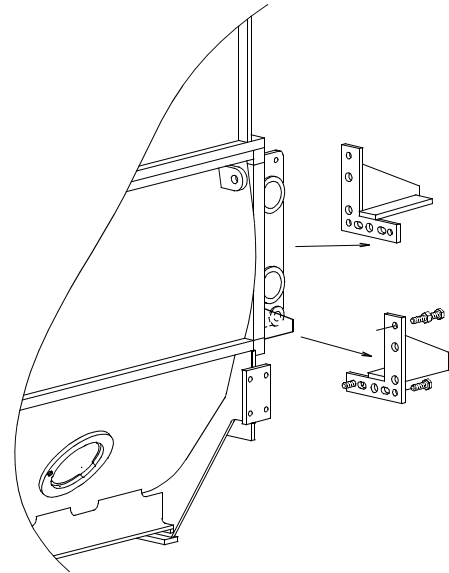
Remember to vent trapped air from the air cooler element.

4.



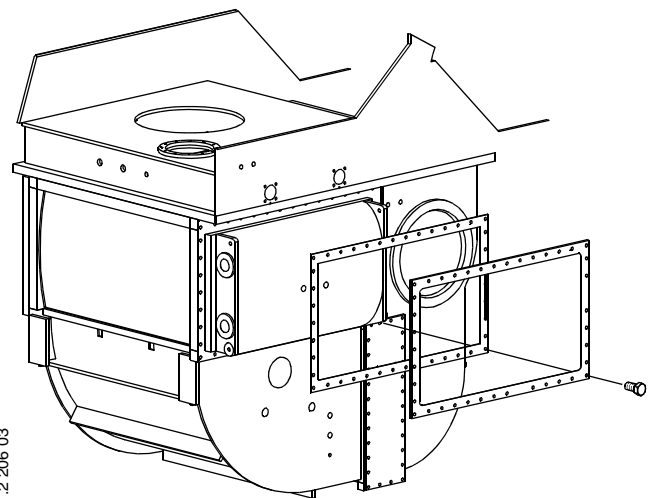
GN910-1.4 228 04

5.



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6.



GN910-1.2 206 03

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME-engines</i>
☒	Shut off control air supply
☒	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit



The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description

The non-return valves are accessed from the scavenge air receiver.

Note!

Access to the scavenge air receiver must **not** be attempted until the air in the receiver is clean.

Main Non-Return Valve

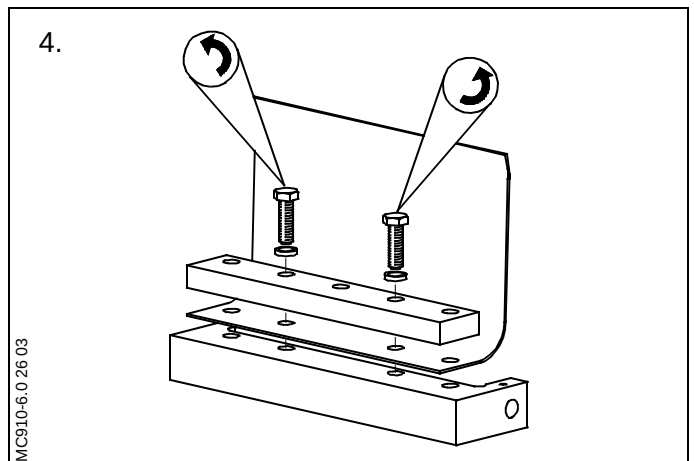
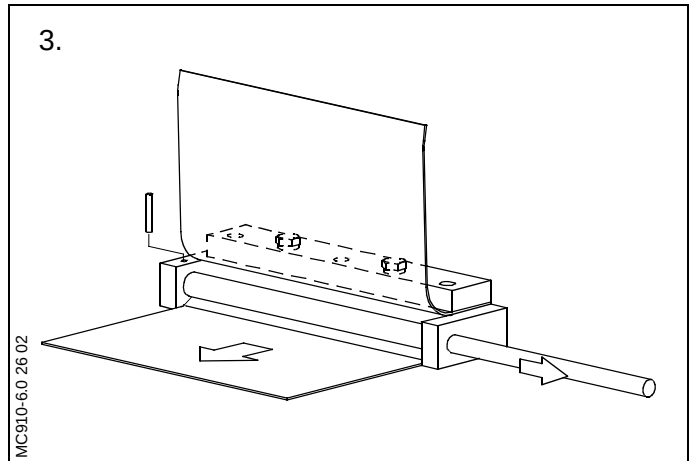
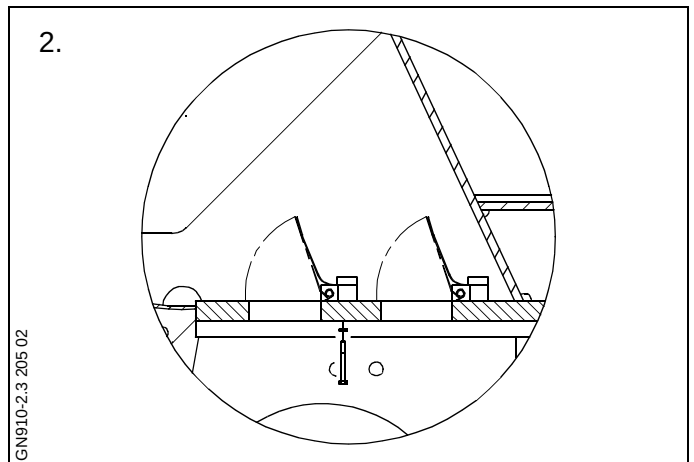
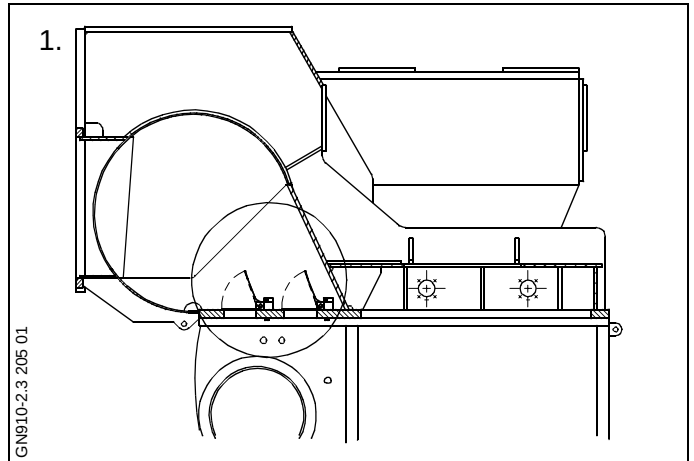
1. Inside the scavenge air receiver, lift up the valve flap of the valve that is to be overhauled.
2. Remove the screws behind the valve flap. Remove the valve from the engine.
3. Remove the spring pin from the valve shaft and knock the valve shaft out of the valve flap.
4. For further dismantling, remove the two screws which hold the clamp and the support together as an assembly.

Clean the valve shaft and the slide bearings inside the valve flaps. Lubricate the valve shaft and the slide bearings with molybdenum disulphide (MoS_2).

5. Assemble the valve

After the overhaul, it must be possible to move the valve flap by the light touch of a finger.

6. Re-mount the valve in the scavenge air receiver.



SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D10-09	Weight of blower unit, complete * up to *) Depending on electric motor type and number of cylinders of engine (size and capacity of auxiliary blower)	800	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description
P91356 P91361		Lifting Tools, etc. Combination Wrenches

When cleaning and inspecting the auxiliary blower, take out the electric motor and the blower wheel as one whole assembly.

1. Disengage the cable connections to the blower motor.

Attach a wire rope with shackle to the lifting mounting on the blower flange, and hook on the tackle suspended from mountings on the gallery bracket.

Remove the screws in the blower wheel flange, and pull the electric motor with flange and blower wheel out of the end cover, using the tackle.

2. Inspect and clean the blower wheel as well as the blower housing.
3. If the whole blower assembly is to be dismounted, the inspection covers on the scavenge air receiver are first to be removed.

Caution!

Make sure that the air in the receiver is clean before making the access.

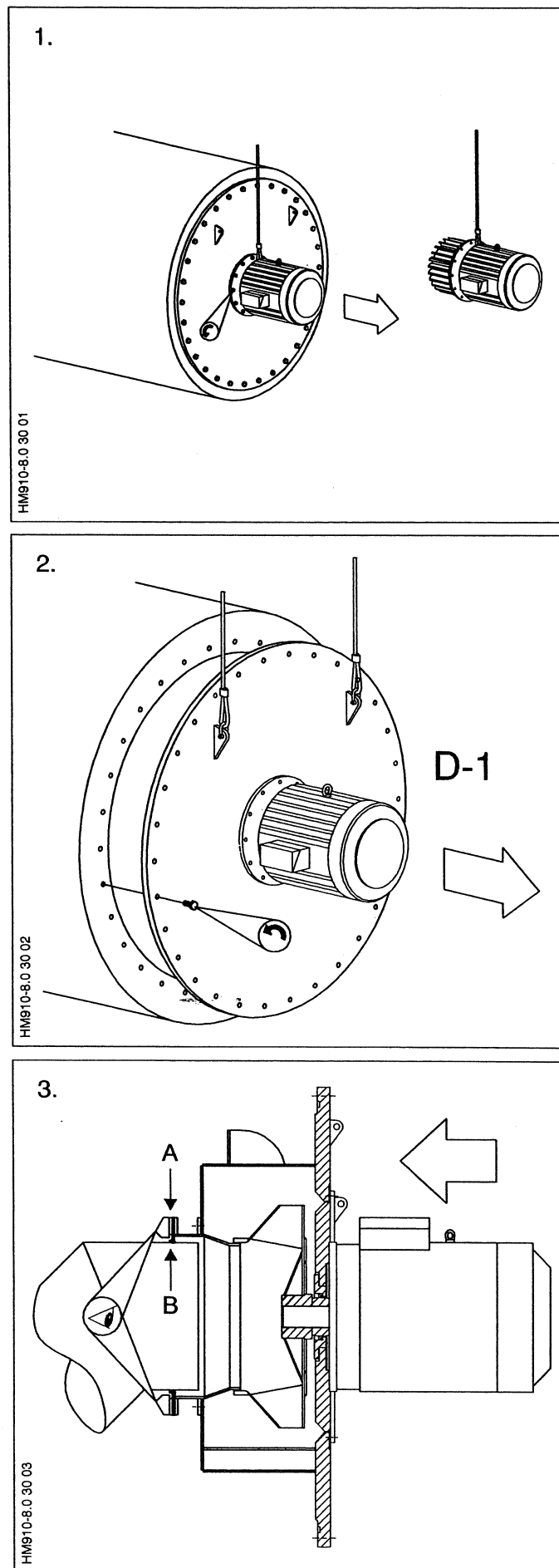
Suspend two tackles from the mountings on the gallery bracket above the end cover.

Attach two wire ropes with shackles to the lifting mountings on the end cover, and hook on the tackles.

Remove all screws holding the end cover. The end cover complete with blower can now be pulled free of the suction pipe by means of the two tackles. While pulling out the end cover with the blower assembly, make sure that the blower housing disengages from the air pipe without deforming it.

4. When mounting the end cover with the complete blower, the parts are to be "guided" into place with care, making sure that the pipe in the receiver "catches" the guide bosses (A) of the blower housing flange.

The gasket (B) constitutes the sealing between blower housing and pipe.



SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☐	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☐	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description
P91361		Combination Wrenches

1. It is important that the automatic non-return valves (butterfly valves) are always able to function easily and unimpeded.

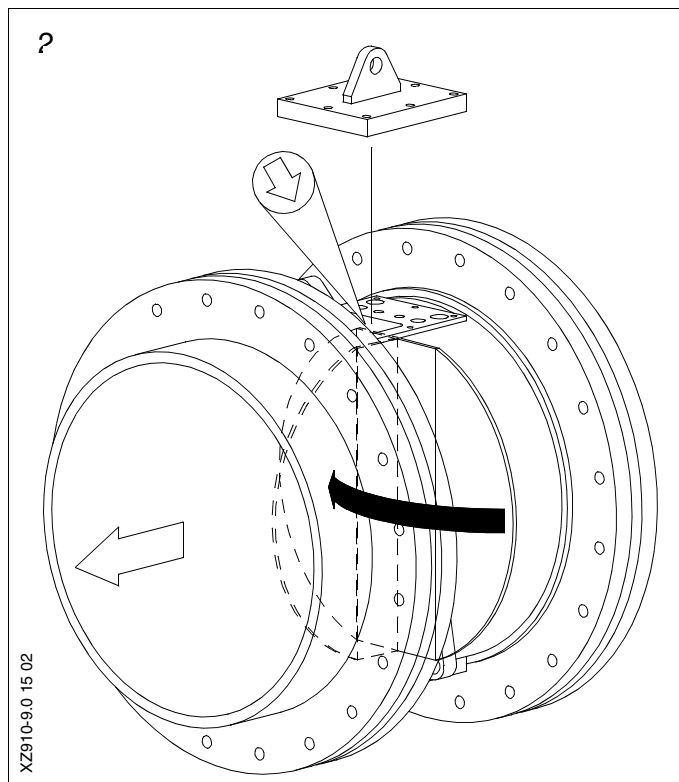
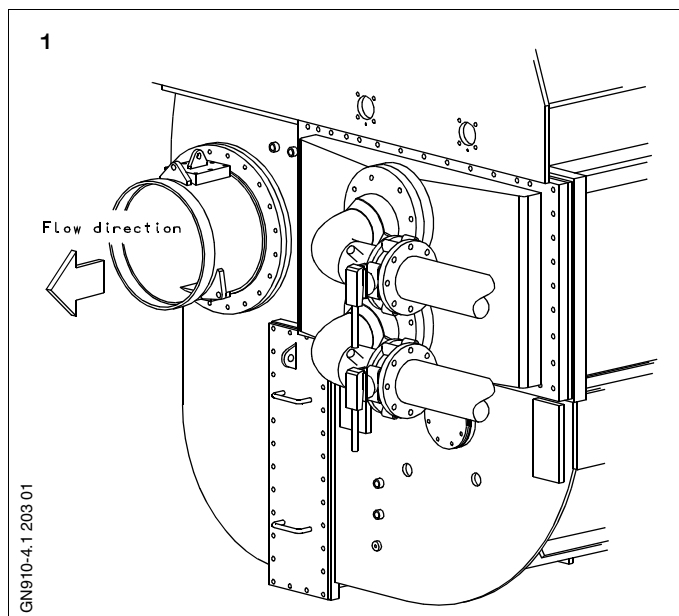
The movability of the valve flaps should therefore be checked at suitable intervals. The valves shall open for the air flow from the scavenge air cooler housings to the auxiliary blowers.

2. To check the valve, remove the cover on the top of the butterfly valve.

Reach down through the opening in the top of the valve and move the valve flaps.

3. If the valve flaps in one of the butterfly valves cannot be moved by a light touch of a fingertip, it must be dismantled and overhauled.

4. Re-mount the cover on the top of the butterfly valve.



SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

- ☐ Stopped engine
- ☐ Shut off starting air supply – *At starting air receiver*
- ☐ Block the main starting valve
- ☐ Shut off starting air distributor/distributing system supply
- ☐ Shut off safety air supply – *Not ME-engines*
- ☐ Shut off control air supply
- ☐ Shut off air supply to exhaust valve – *Only with stopped lubricating oil pumps*
- ☐ Engage turning gear
- ☐ Shut off cooling water
- ☐ Shut off fuel oil
- ☐ Stop lubricating oil supply
- ☐ Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description

The dry cleaning method employs compressed air to blow dry, solid, granules through the turbocharger, thus removing most of the deposits which may have formed on the nozzle vanes and turbine blades.

On account of their hardness, nut-shells, or broken or artificially shaped pieces of activated charcoal with a size of between 1.0 and max. 2.8 mm are particularly suited as blasting agents.

For exact information about cleaning agent size, see *manufacturers documentation*.

Note!
We do not recommend the use of rice or grain as cleaning materials, as these may possibly stick in the exhaust gas boiler.

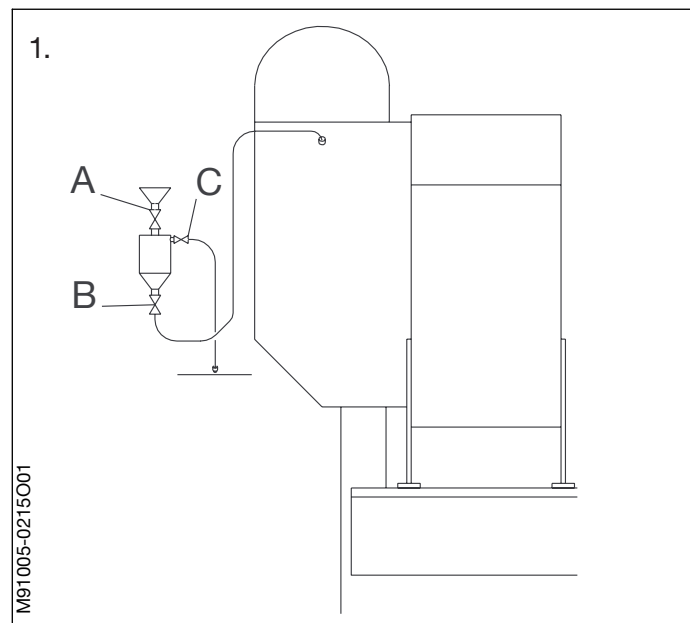
It is not always possible to remove thick deposits with this dry cleaning method.

Therefore, in order to prevent the build-up of thick deposits, the turbine must be cleaned after every 24 to 50 hours of operation.

Generally, turbocharger cleaning can be carried out without, or with very little, reduction of the engine load.

More detailed information regarding the amount of granules to be used is stated on the instruction plate located close to the turbocharger.

Note!
<i>Regarding water washing of turbocharger turbine, see turbine manufacturer's instructions. See also Volume I, Chapter 706.</i>

**Cleaning Procedure:**

1. Carry out cleaning for every 24 to 50 hours of operation, based on observations.
2. Clean the turbocharger at an engine load between 50%-90%. Do not clean below half load.
3. Close valve **A**.
4. Open valves **B** and **C**, to blow out any deposits and/or condensate in the connecting pipe.

After about two minutes, close valves **B** and **C**.

5. Slowly open valve **A** to relieve the pressure in the tank.
6. Fill the tank with the quantity of granules specified in the table.
7. Close valve **A**.
8. Open valves **B** and **C**, to blow-in the granules. After one to two minutes, close valves **B** and **C**.
9. Slowly open valve **A** to relieve the pressure in the tank.

2.

TC type	Amount dm ³
TCA 33	0.5
TCA 44	0.5
TCA 55	1.0
TCA 66	1.5
TCA 77	2.0
TCA 88	2.5
TCA 99	3.0
TPL 61	0.5
TPL 65	0.6
TPL 69	0.9
TPL 73	1.2
TPL 77	1.5
TPL 80	2.0
TPL 85	3.0
MET 33	0.4
MET 42	0.7
MET 53	1.6
MET 60	2.1
MET 66	2.6
MET 71	3.0
MET 83	3.5
MET 90	3.5

Caution!

1. The drain openings in the gas casings must remain closed while dry cleaning the turbine.

2. It is possible that during dry cleaning of the turbine, some of the blown-in solid particles or sparks will escape through the funnel.

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☐	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☐	Engage turning gear
☒	Shut off cooling water
☐	Shut off fuel oil
☐	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D10-11	Water mist catcher	100	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

Plate	Item No.	Description
P91356		Lifting Tools, Etc.

1. Normally it is not necessary to remove the water mist catcher from the scavenge air cooler housing. If, however, the water mist catcher has been fouled or damaged, it must be dismantled for cleaning or repair.
2. Dismantle the rectangular covers from both the front and aft ends of the scavenge air cooler housing.
3. Loosen the screws at both ends of the water mist catcher.

Note!

Note the position of the water mist catcher. During mounting, the water mist catcher must be re-mounted in the same position.

4. Pull or push in the water mist catcher half-way out and mount an eye bolt in the top of the water mist catcher.

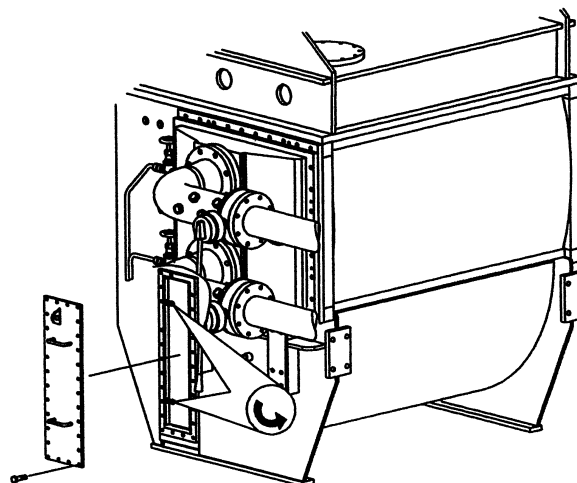
Hook a tackle on to the eye bolt and lift the water mist catcher away. Land it on a couple of wooden planks.

Note!

Take care not to damage the water mist catcher during the lift.

5. Clean the water mist catcher with the same cleaning fluid as used for the scavenge air cooler element.

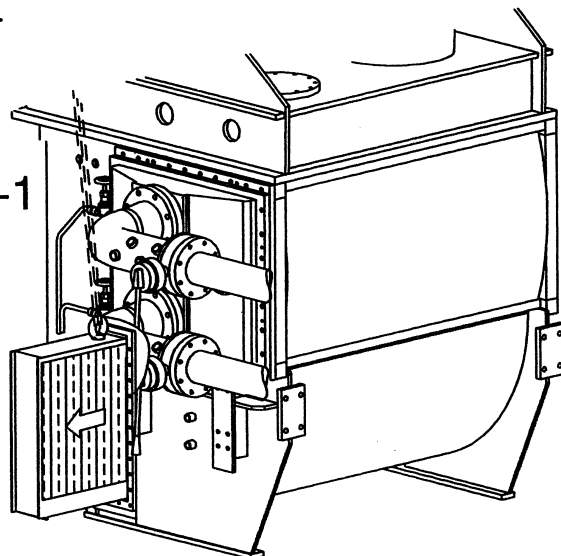
2.



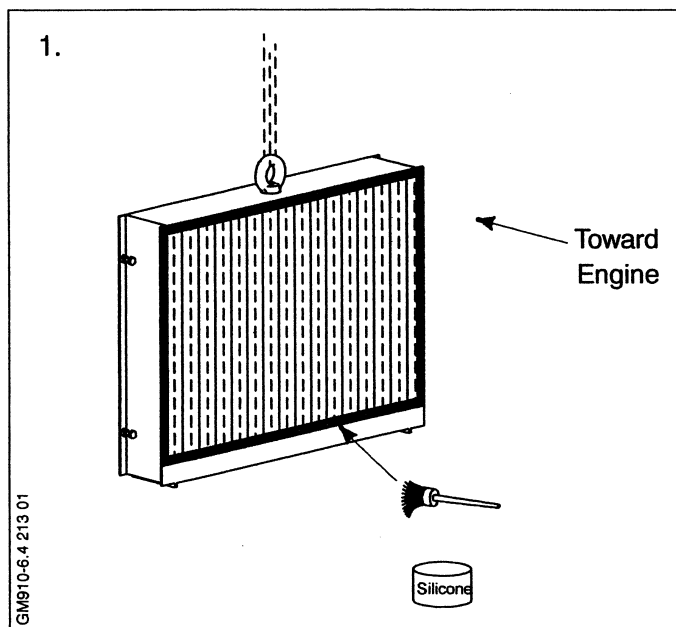
GM910-6.2 213 02

4.

D-1



GM910-6.2 213 04

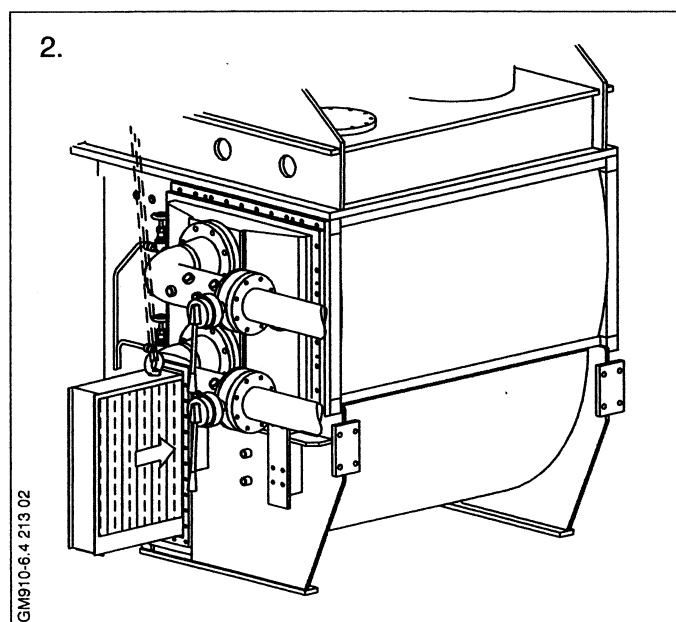


1. Apply a thin layer of silicone paste to the frame of the water mist catcher on the side which will face away from the engine when the water mist catcher is mounted (the side opposite to the screws).
2. Slide the water mist catcher into the scavenge air cooler housing.

Note!

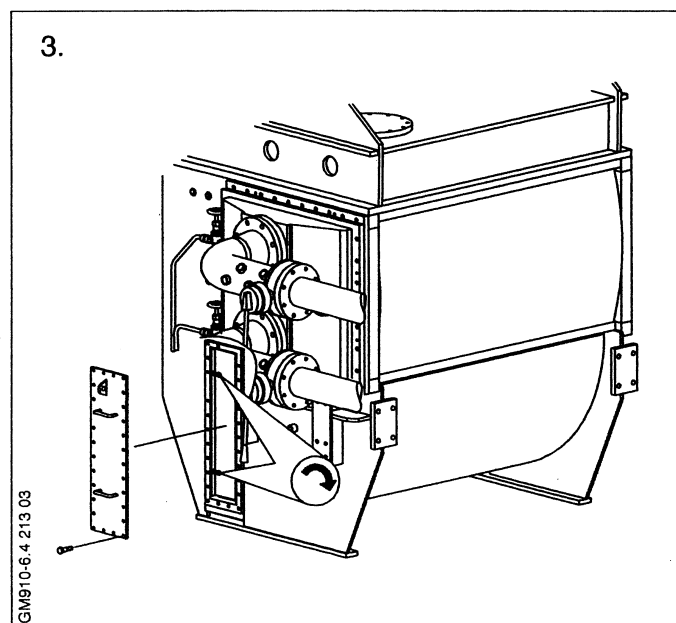
Make sure to mount the water mist catcher correctly. The screws of the water mist catcher must face the exhaust side of the engine.

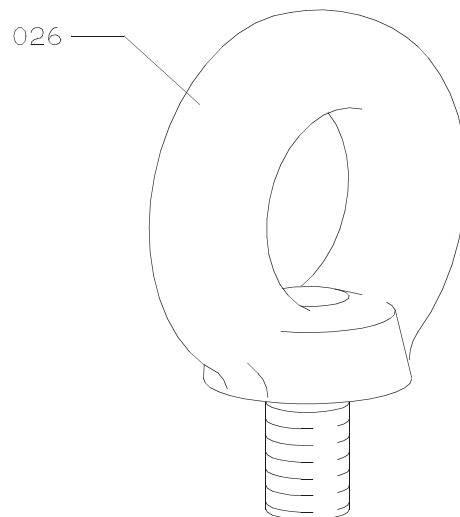
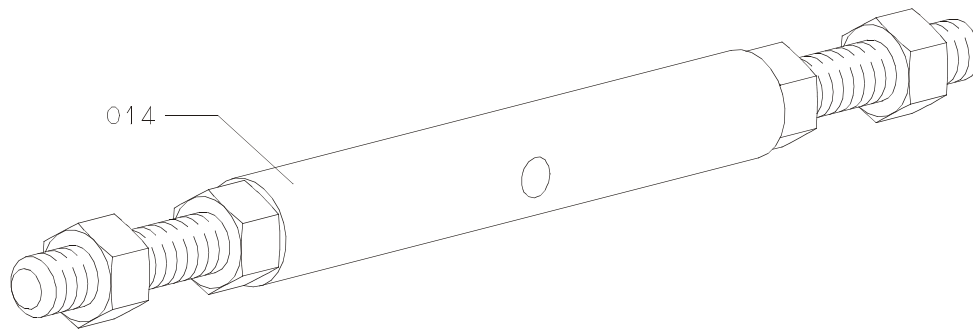
When the water mist catcher is half-way in, remove the tackle and the eye bolt, and push the water mist catcher fully into the scavenge air cooler housing.



3. Tighten the screws at both ends of the water mist catcher.

Apply a thin layer of silicone paste on the frame and mount the rectangular covers on the front and aft ends of the scavenge air cooler housing.

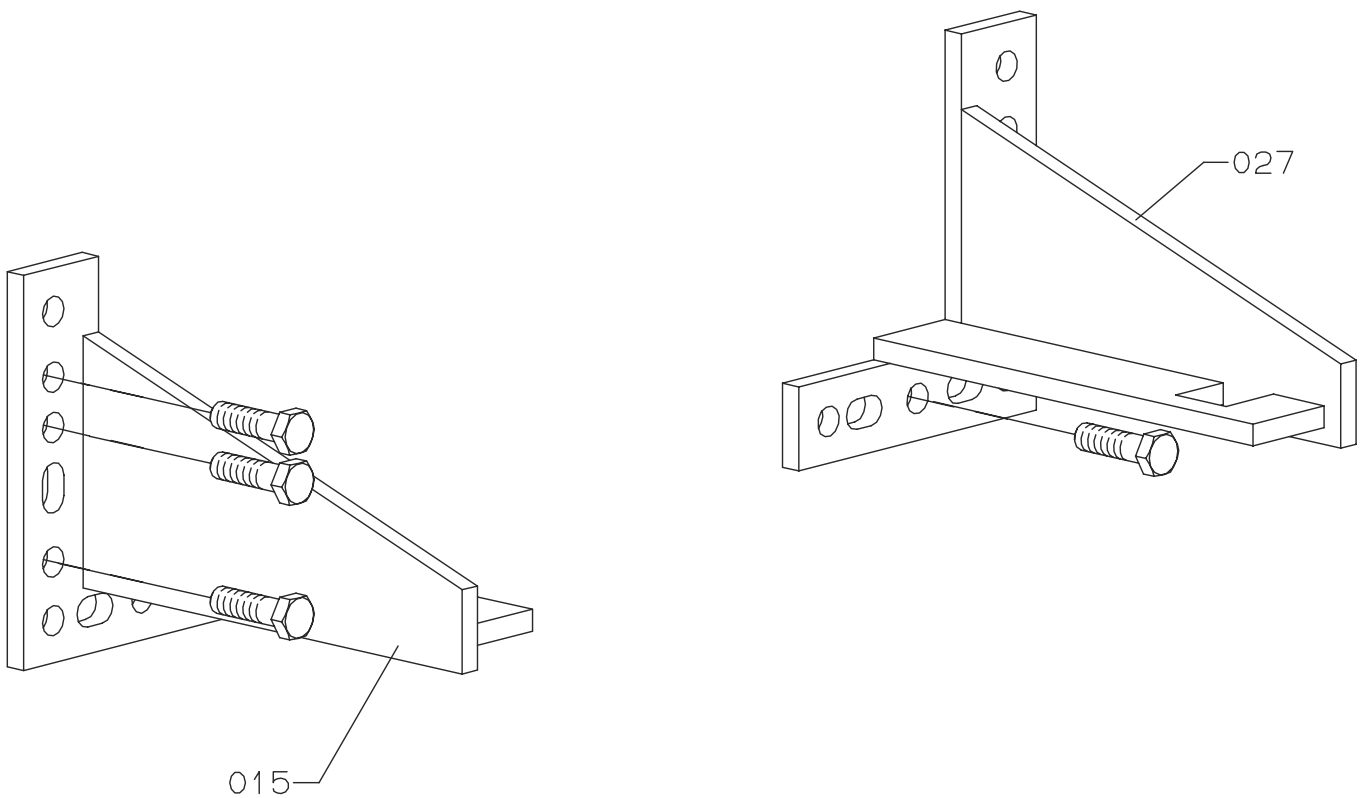




Item No.	Item Description
014	Pulling tool
026	Lifting eye bolt

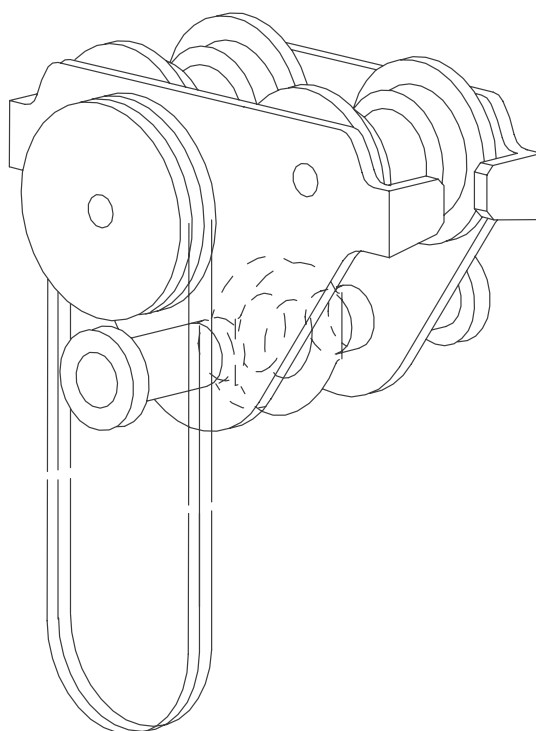
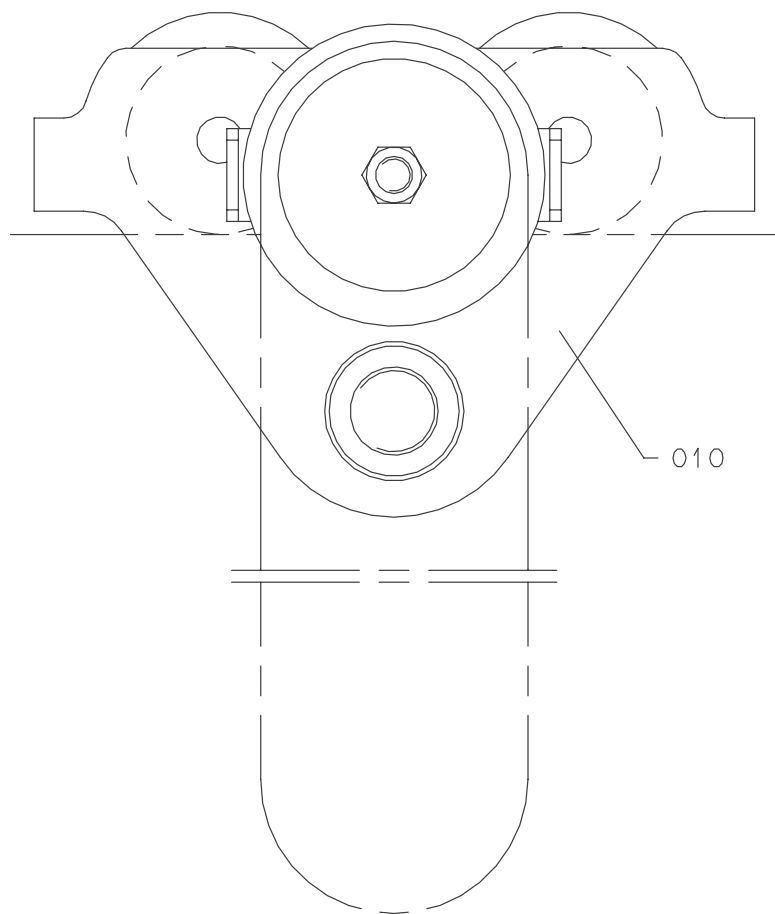
Item No.	Item Description

Fore end





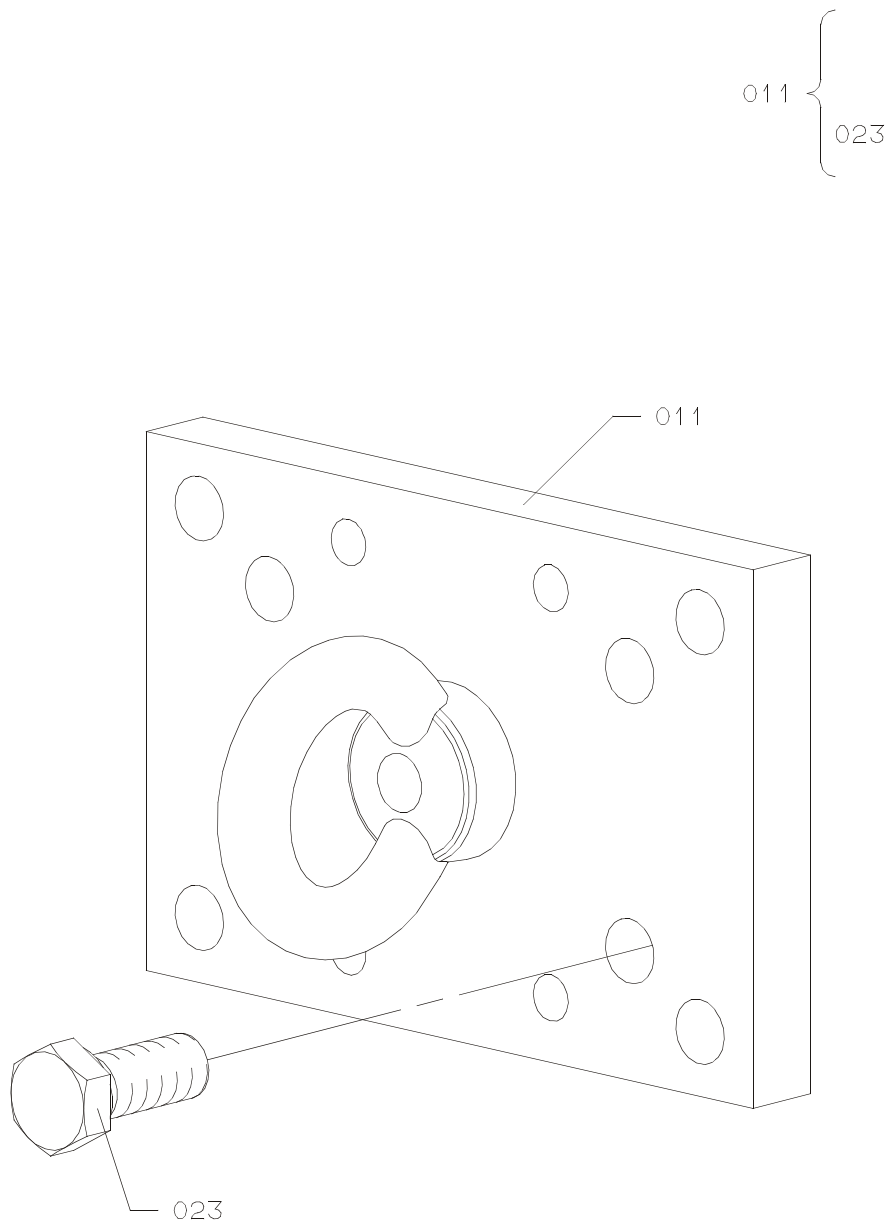
Item No.	Item Description
015	Guide rail, air cooler fore end
027	Guide rail, air cooler fore end





Item No.	Item Description
010	Travelling trolley* Note: * When ordering this tool, please state load and track width.

Item No.	Item Description



Item No.	Item Description
011	Lifting attachment air cooler fore end, complete
023	Screw

Item No.	Item Description